Statistical Case Studies, Project 1 Part 1, Code

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Executive Summary

Data

- Description of data

Filtering and Cleaning

Table 1: Table 1 — quality codes and their frequency in the data

Code number	Code name	Abbreviated code name	Count	Percentage
50	Good	G	335,515	95.70%
100	Estimated	E	14,915	4.25%
140	Provisional	Prov	0	0.00%
150	Suspect	S	0	0.00%
200	Unchecked (imported from legacy database)	V	0	0.00%
254	Unchecked	U	162	0.05%

- Justify why kept 100 and above but not 254

Exploratory Analysis

Throughout the report, we add three time variables to support our analysis: the month, the season, and the water year. The water year runs from September to August and is labelled by the year it ends in, which keeps each winter's high flows together in the same year.

Main time series and summary table

We first construct the hydrograph for Don at Parkhill over our chosen period.

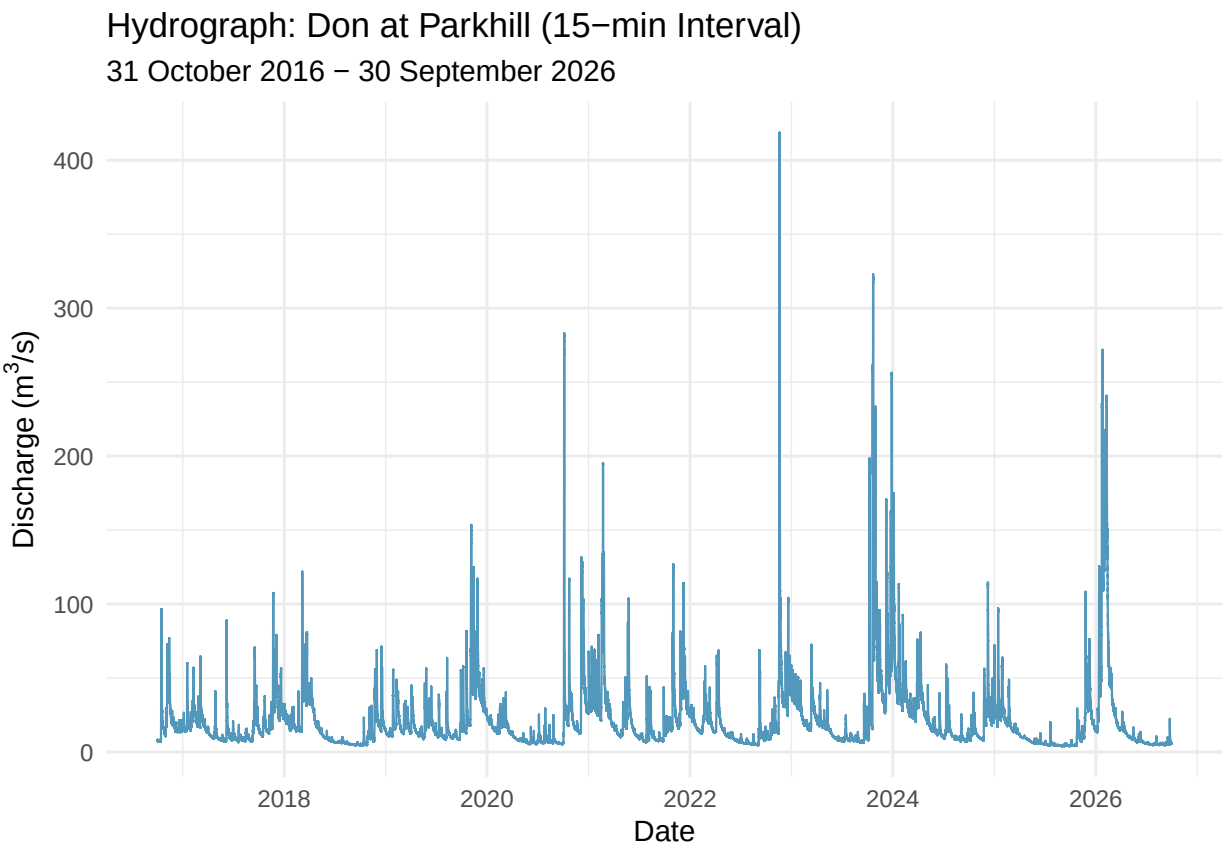
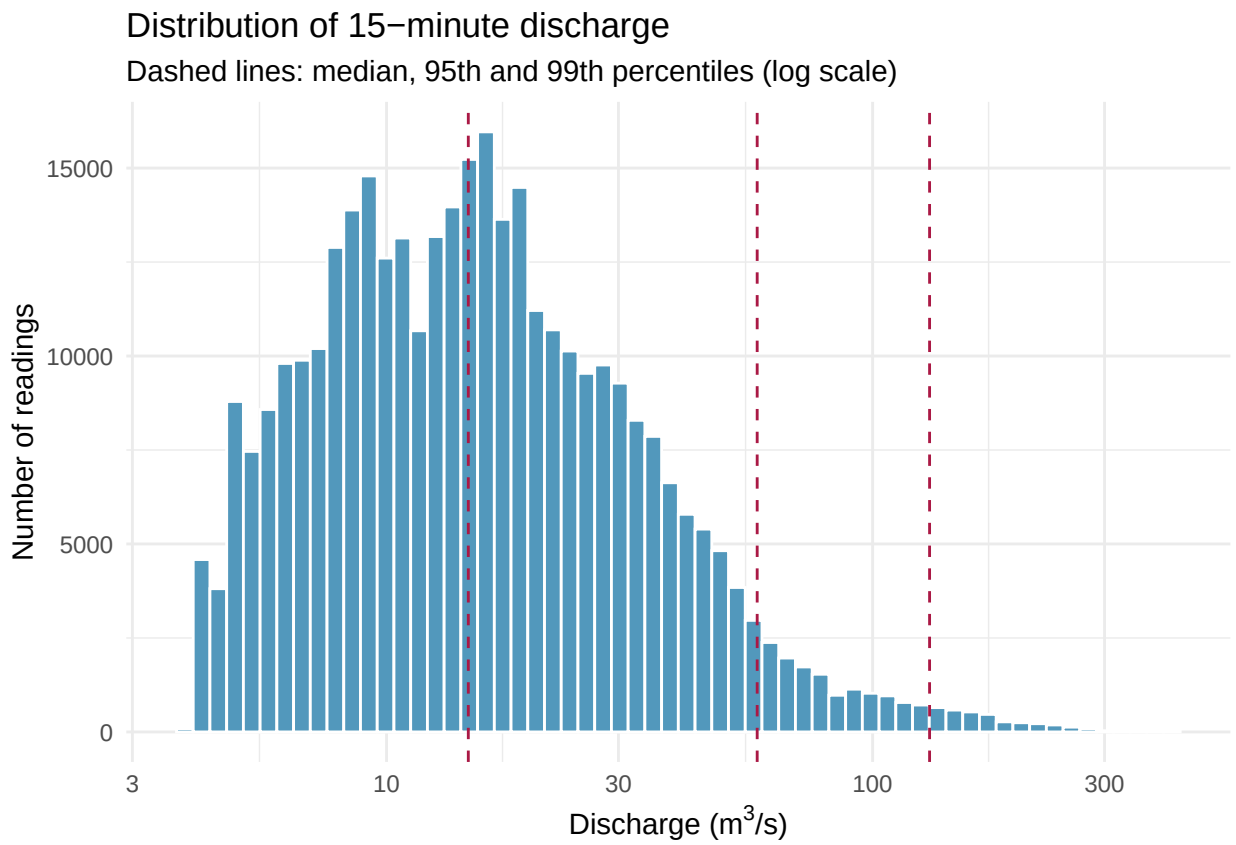


Table 2: Summary statistics for 15-minute discharge (m^3/s)

Statistic	Value
Mean	21.65
Standard deviation	24.38
Minimum	3.90
25th percentile	8.74
Median	14.73
75th percentile	25.44
95th percentile	57.88
99th percentile	130.96
Maximum	418.68



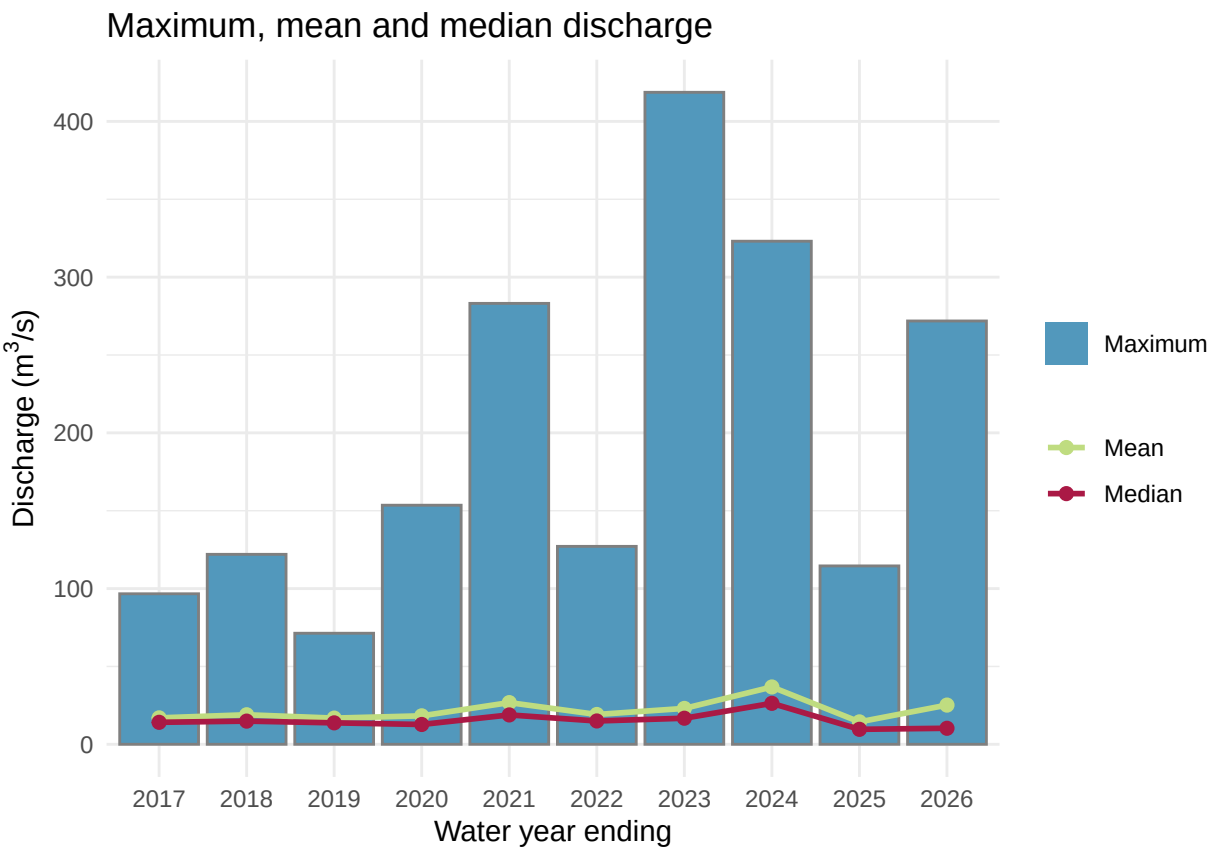
We observe that discharge is low for most of the year, with a smaller number of large peaks. The strong-right skew of the distribution confirms this, with a median discharge of 14.91 m^3/s, and a 99th percentile of 130.96 m^3/s.

Yearly trends

From figure ..., we also observe an upward trend in discharge, and larger maximum flows, over the time period.

Table 3: Discharge by water year (m^3/s)

Water year	Mean	Median	Maximum	Date of maximum
2017	16.99	14.12	96.65	16 Oct 2016
2018	18.97	14.89	121.98	07 Mar 2018
2019	16.83	13.77	71.30	16 Dec 2018
2020	18.29	12.73	153.53	05 Nov 2019
2021	26.77	18.86	283.14	05 Oct 2020
2022	19.16	14.98	127.10	01 Nov 2021
2023	23.05	16.75	418.68	19 Nov 2022
2024	36.82	26.31	323.07	22 Oct 2023
2025	14.39	9.62	114.55	08 Dec 2024
2026	25.18	10.29	271.84	25 Jan 2026


Maximum, mean and median discharge
Discharge (m3/s)
400
300
200
100
0
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
Water year ending
Maximum
Mean
Median

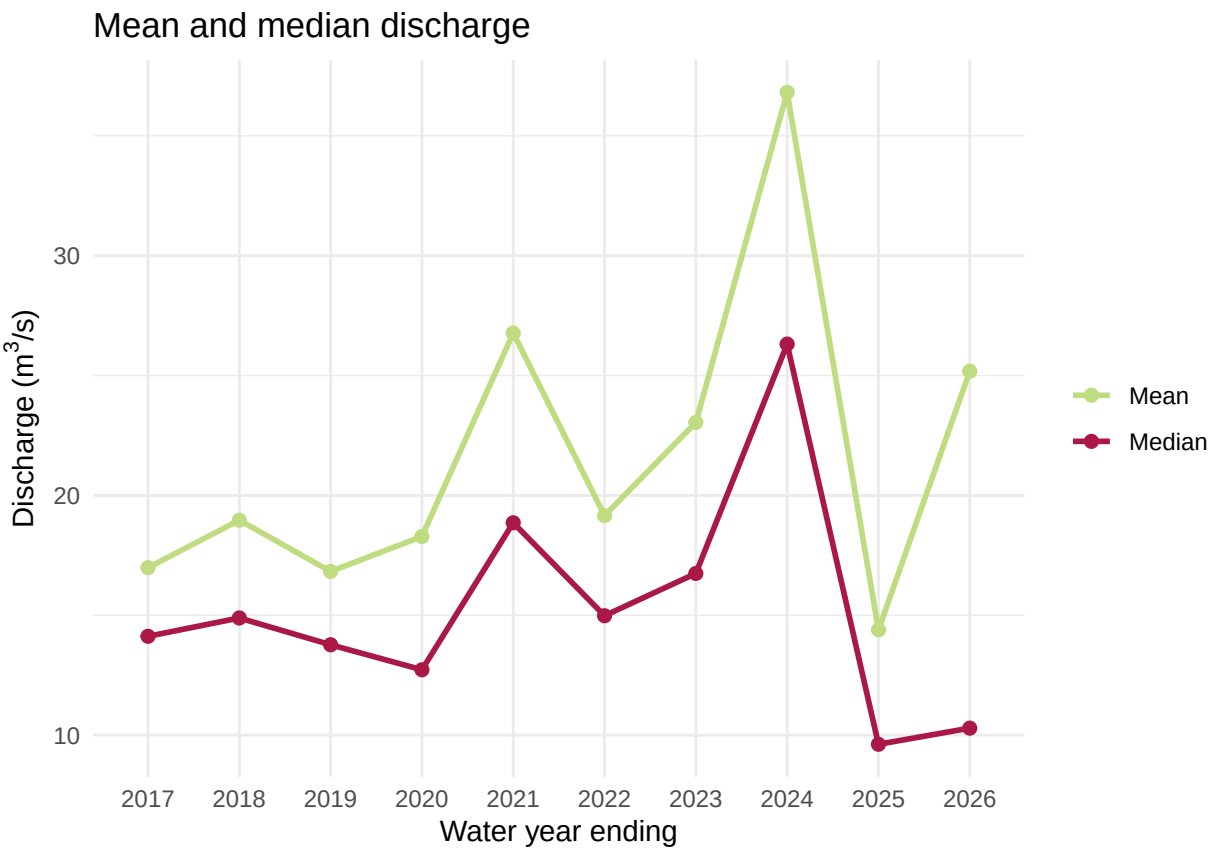


Figure ... provides evidence of this observation, the mean discharge trending upwards from the water year ending in 2017 at ..., peaking in 2024 at .... The maximum discharge follows a similar trend, with both falling sharply in 2025 before rising again.

Exploring seasonal trends overall

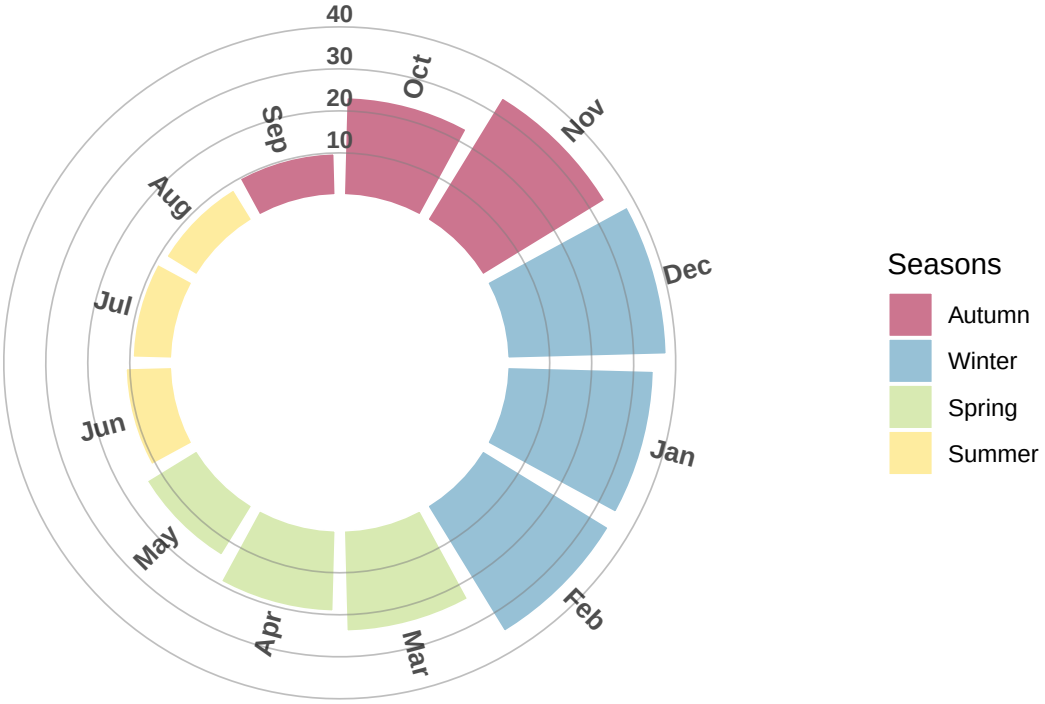
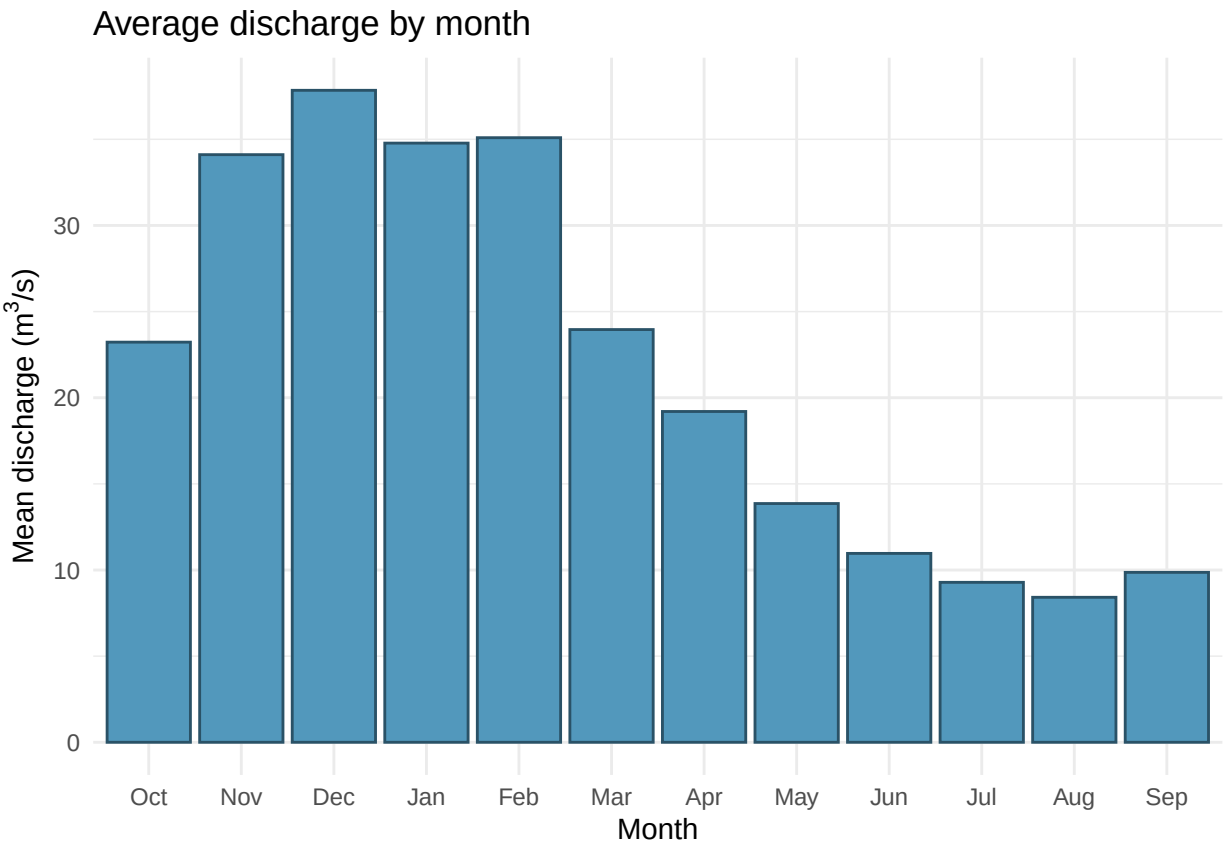
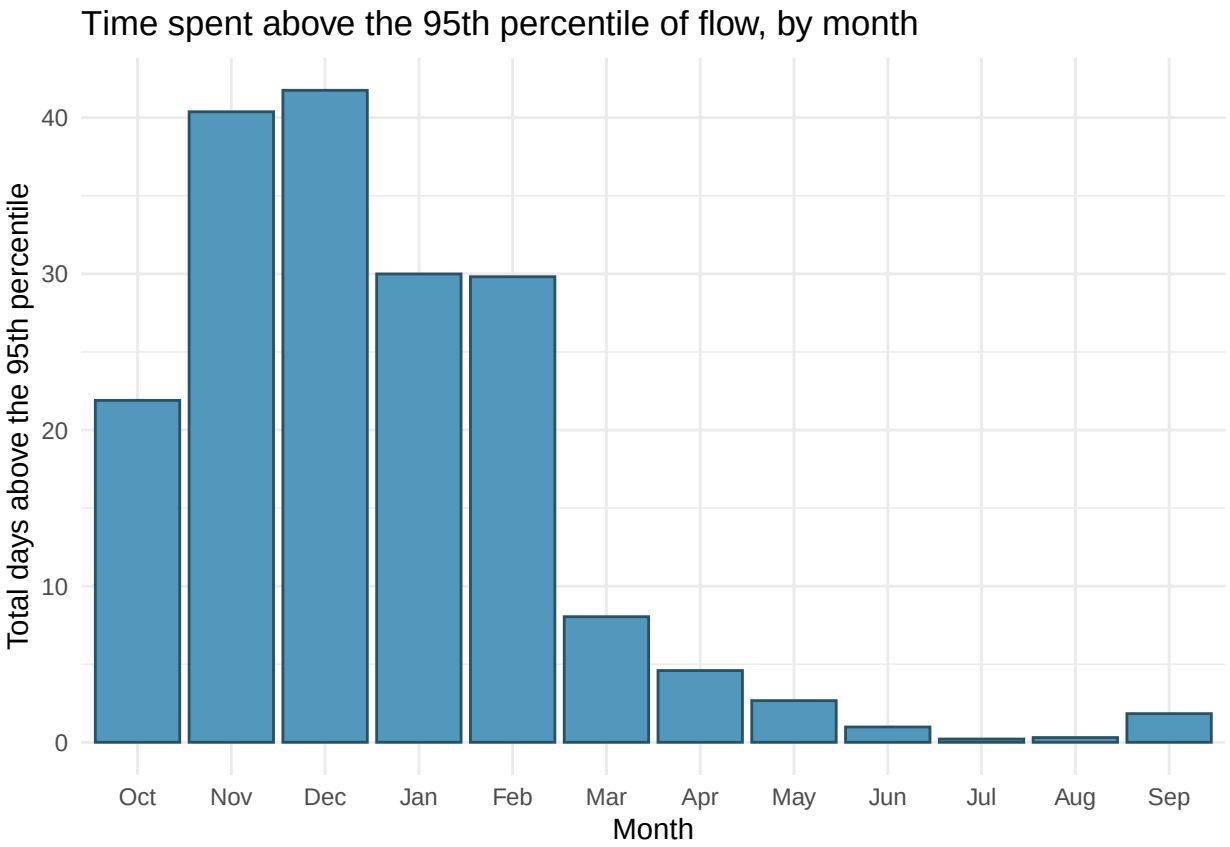
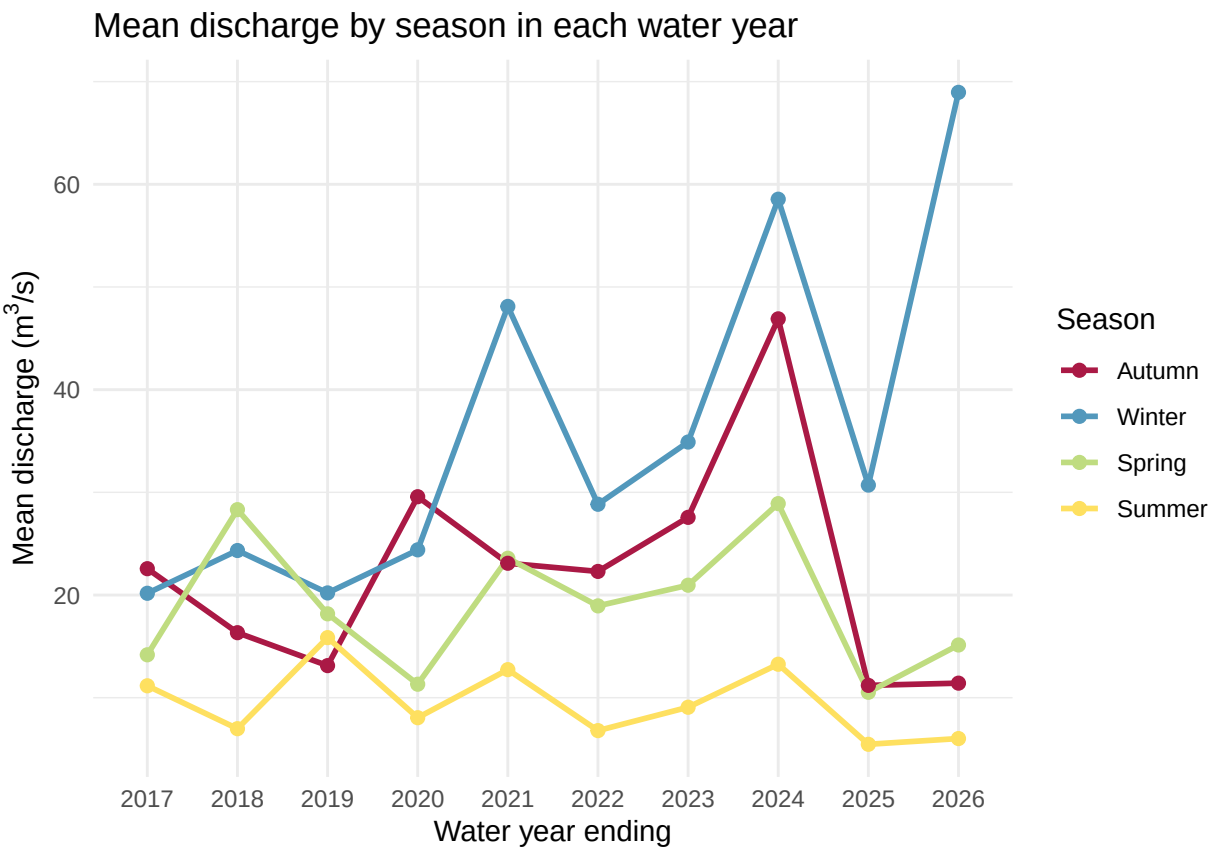


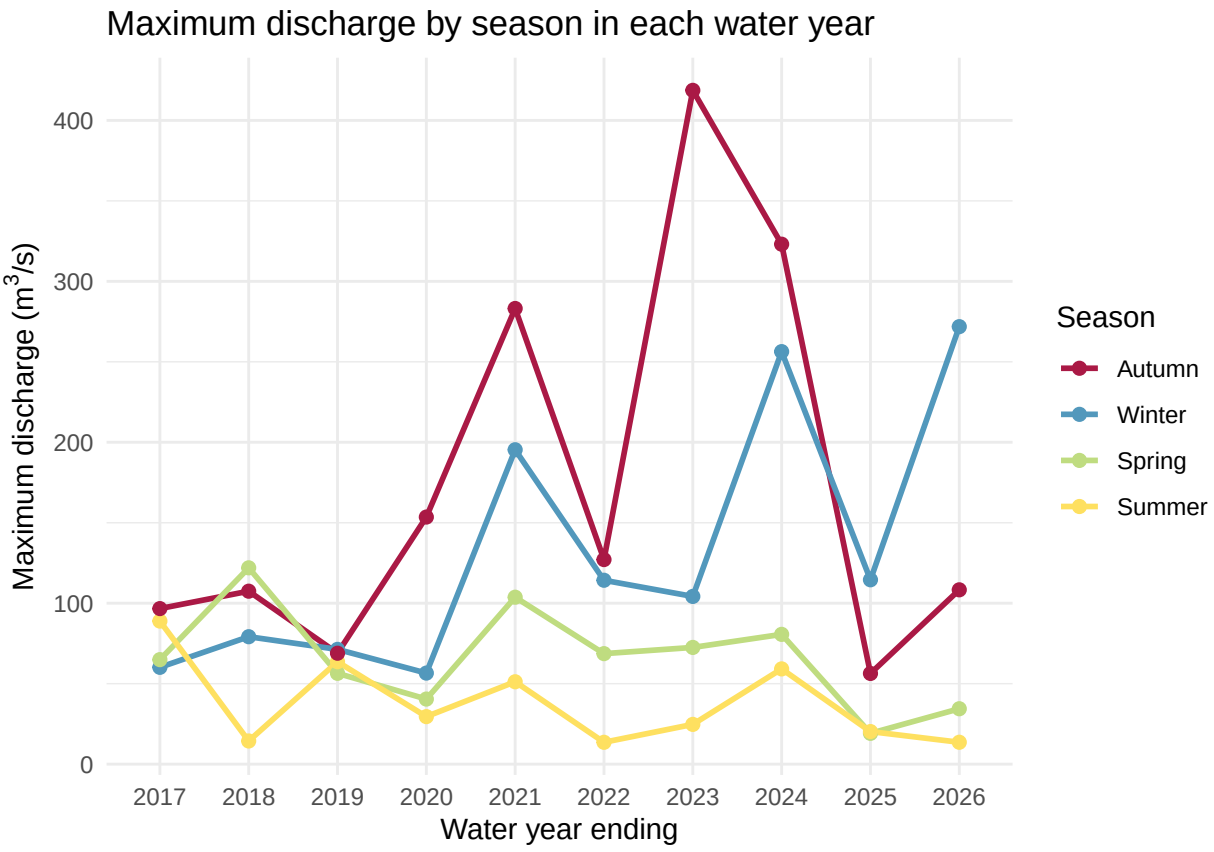
Table 4: Discharge by month (m^3/s)

Month	Mean	Median	95th percentile	Maximum
Oct	23.22	14.49	77.46	323.07
Nov	34.11	22.40	87.85	418.68
Dec	37.84	30.60	90.53	256.33
Jan	34.78	24.78	97.58	271.84
Feb	35.09	23.90	122.03	240.84
Mar	23.95	20.55	44.76	121.98
Apr	19.20	15.31	41.15	80.63
May	13.86	11.14	29.56	103.72
Jun	10.96	8.85	22.97	88.89
Jul	9.28	7.75	18.49	59.17
Aug	8.42	7.76	14.83	63.62
Sep	9.86	7.46	25.34	70.53

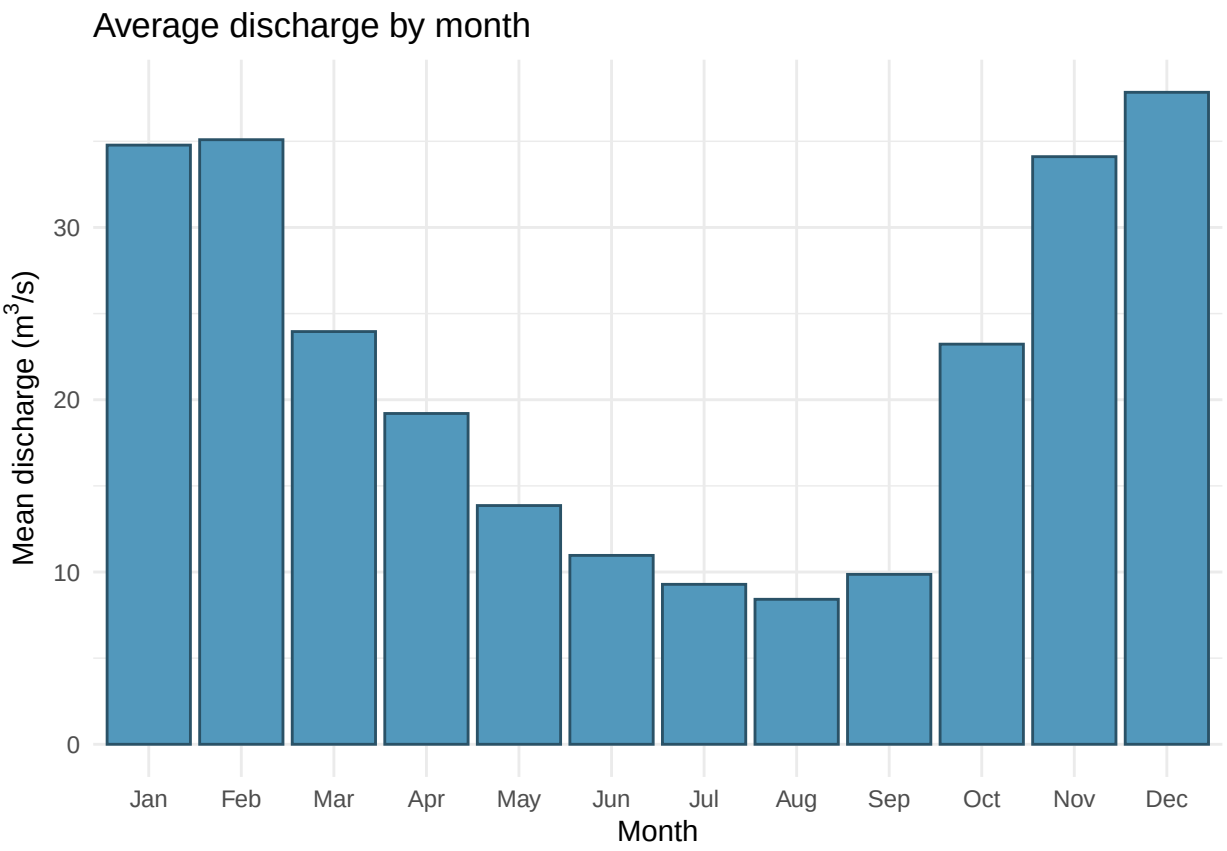

Average discharge by month
Mean discharge (m3/s)
0
10
20
30
Oct
Nov
Dec
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Month


Time spent above the 95th percentile of flow, by month
Total days above the 95th percentile
0
10
20
30
40
Oct
Nov
Dec
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Month


Mean discharge by season in each water year
Mean discharge (m3/s)
60
40
20
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
Water year ending
Season
Autumn
Winter
Spring
Summer



- Other seasonality options.


Average discharge by month
Mean discharge (m3/s)
0
10
20
30
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Month

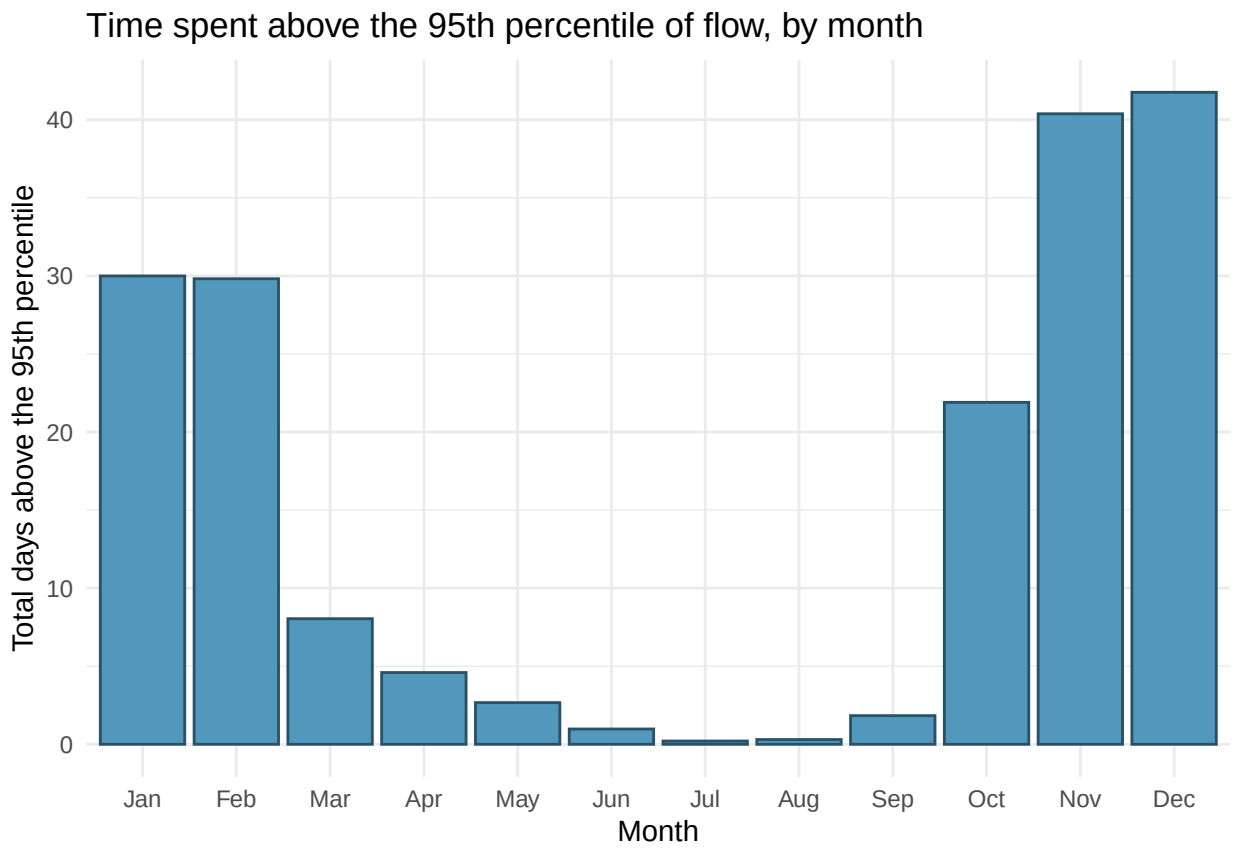
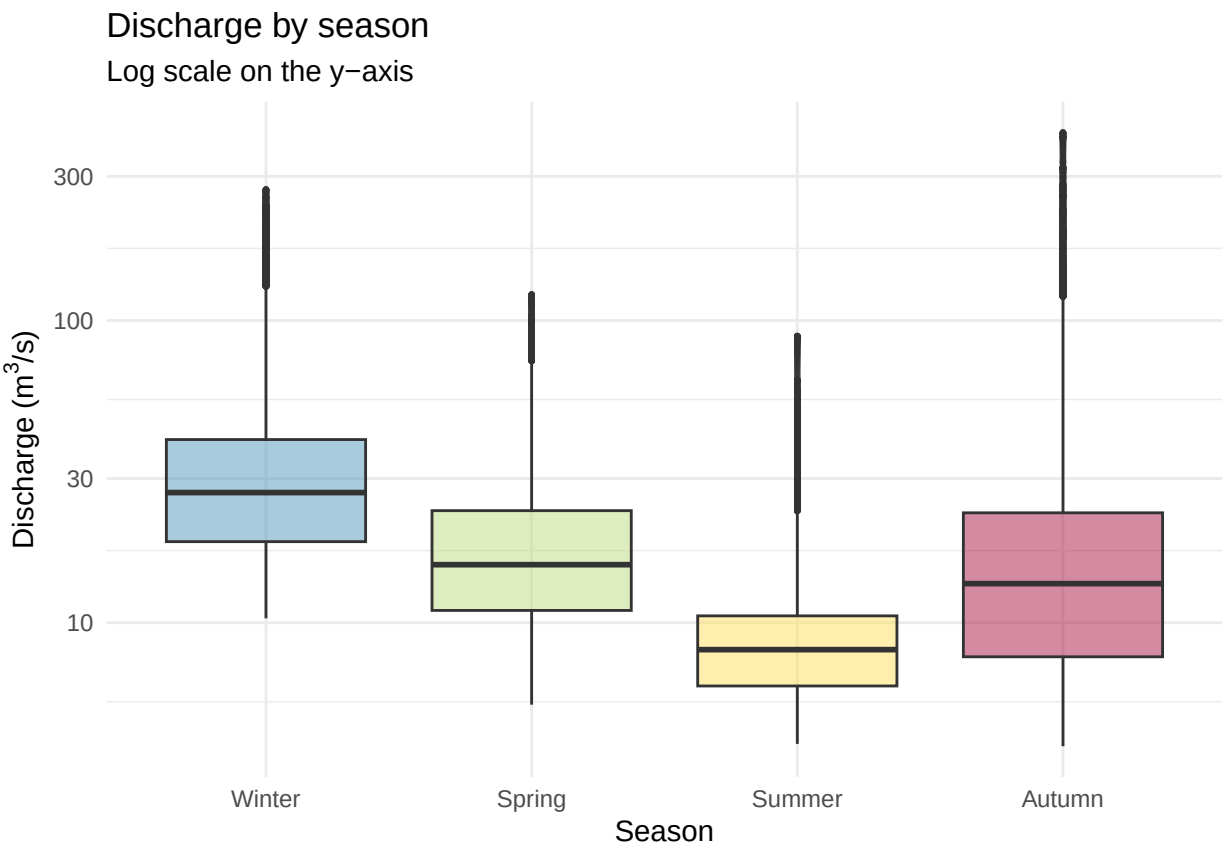


Table 5: Discharge by season (m^3/s)

Season	Mean	Median	95th percentile	Maximum
Winter	35.93	26.95	98.58	271.84
Spring	19.00	15.56	40.43	121.98
Summer	9.54	8.14	20.08	88.89
Autumn	22.41	13.47	69.48	418.68


Discharge by season
Log scale on the y−axis
300
100
30
10
Discharge (m3/s)
Winter
Spring
Summer
Autumn
Season

Threshold Selection

The Peaks-Over-Threshold (POT) uses a threshold to extracts the distinct peak flows for extreme events. The threshold is typically selected from three main methods:

1. 3-5 per year method
2. 95th Percentile method
3. Mean residual life threshold analysis

The threshold defines what is considered an extreme event, hence to verify we have captured all events, a comparison of methods is crucial. The 3-5 peaks per year method can be implemented both visually and mathematically, manually we could count peaks and draw a line, however this is less efficient and accurate. Instead, mathematically we have calculated all of the possible thresholds in which 3-5 extreme events occur, Figure ... below illustrates this with the green region.

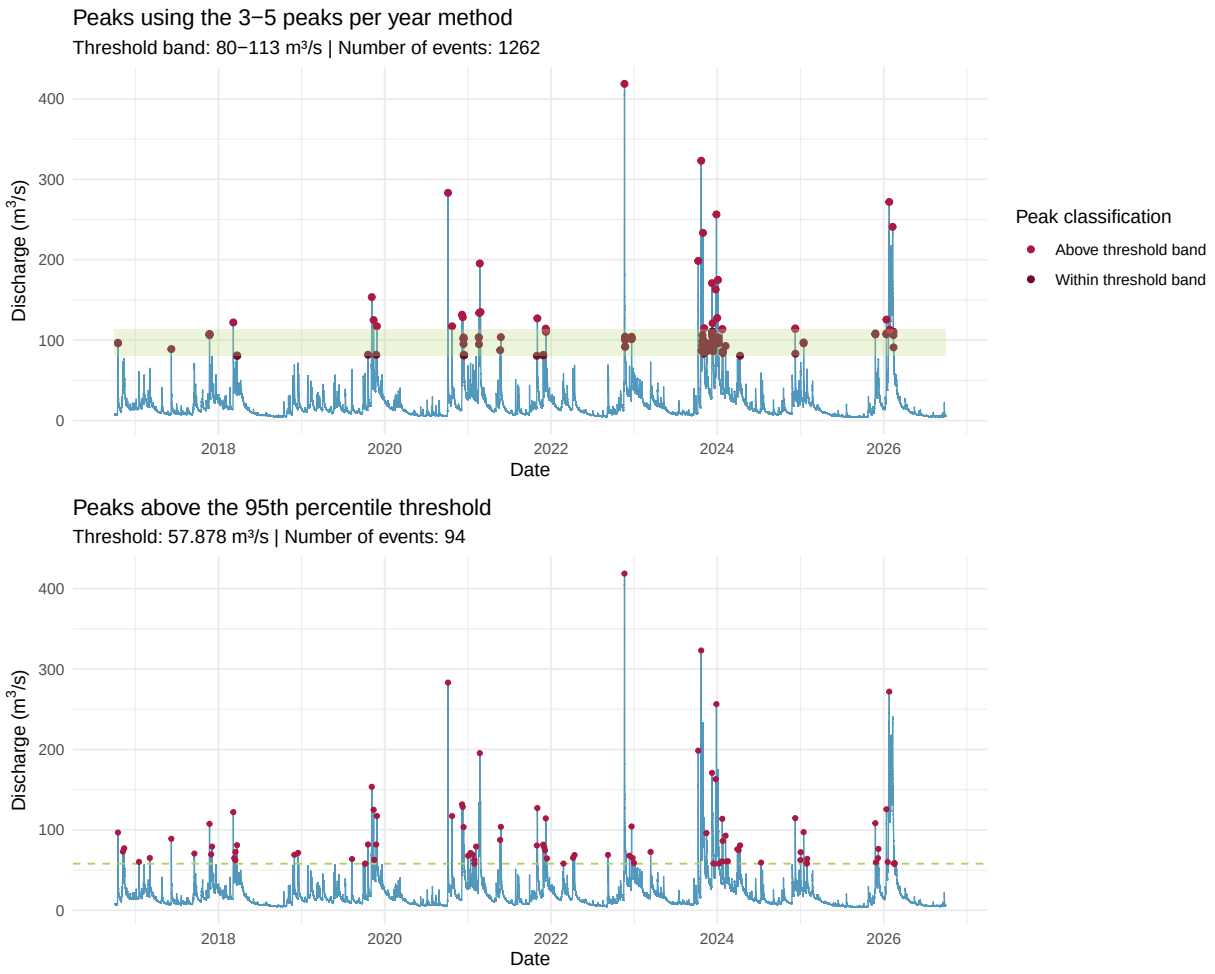


Figure 1: Hydrographs showing Threshold for each Method

The 95th quantile method, creates a threshold line such that 5% of all events are considered extreme events. This is shown on the figure above.

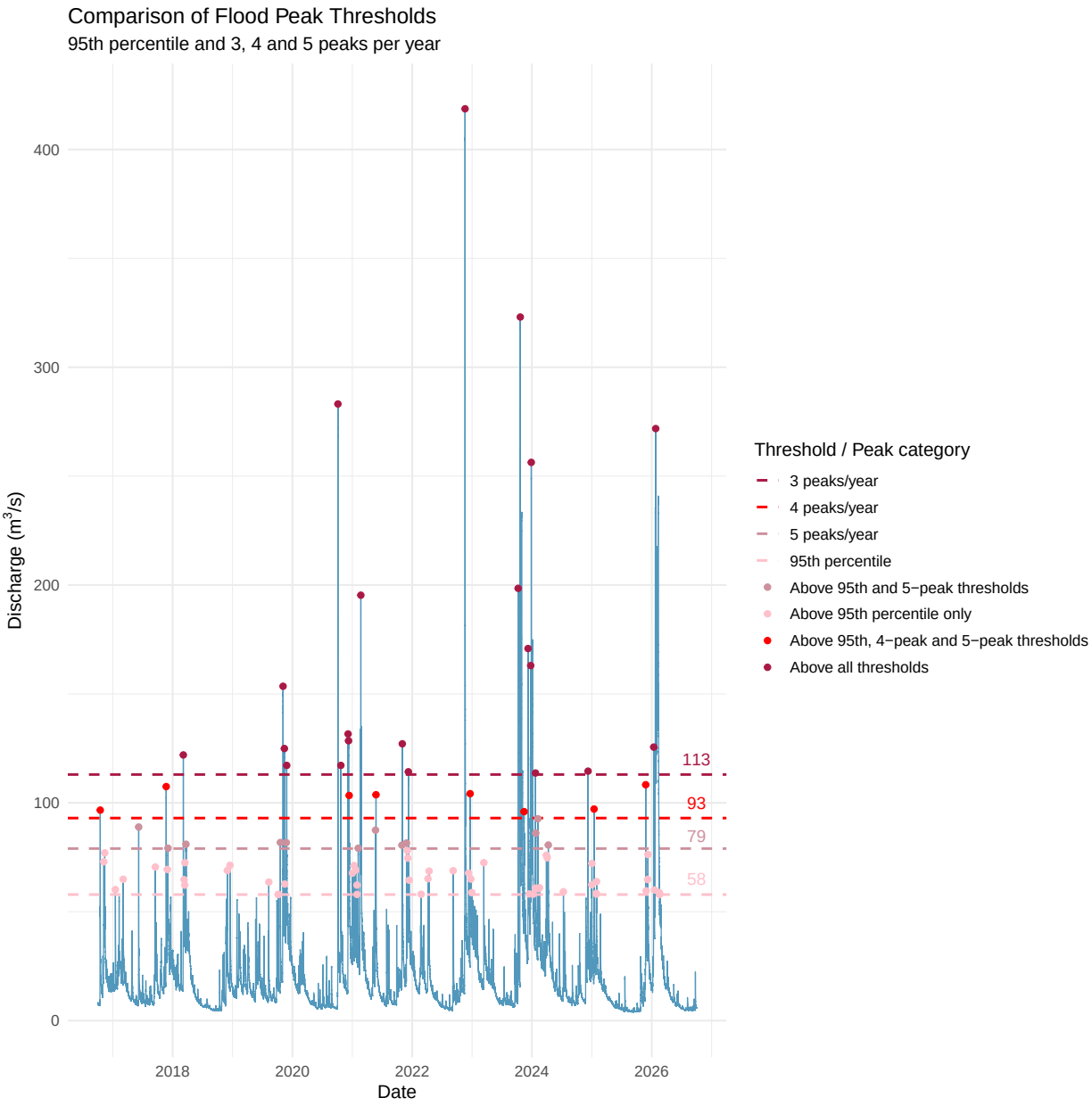


Figure 2: A Comparision of Threshold Methods

The figure above is a comparison of the two methods. Prior to independence analysis, a practical approach is to choose an initial threshold that is likely to be too low, so that when the distinct extreme events are selected, all the major peaks are being analysed, and subsequently the threshold can be raised. The 95th threshold allows for a significant more amount of peaks to be considered, hence we choose a $u = 79$ (5 peaks/year) as an initial threshold. Robson and Reed (1999) verify this "Setting the abstraction threshold low enough to produce an average of about five peaks a year means that, for analytical purposes, there is the flexibility to raise the threshold to exclude the smaller floods."

The Generalized Pareto Distribution (GPD) is used to analyse the initial threshold ($u = 79$) and assess whether the asymptotic approximation is reasonable.
From the mean residual life plot, the relationship appears approximately linear from around $u = 79$ over a range of thresholds.
The parameter stability plots provide further evidence for threshold selection, while the diagnostic plots assess the fitted GPD model.

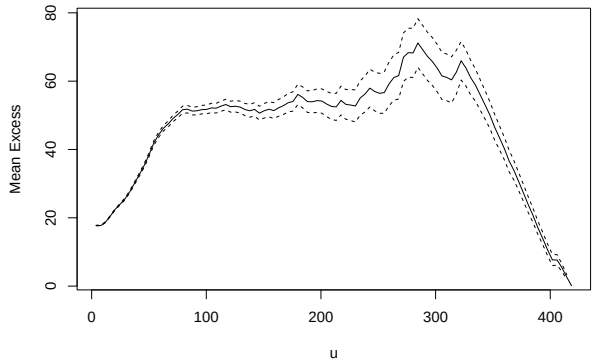


Figure 3: MRL Plot

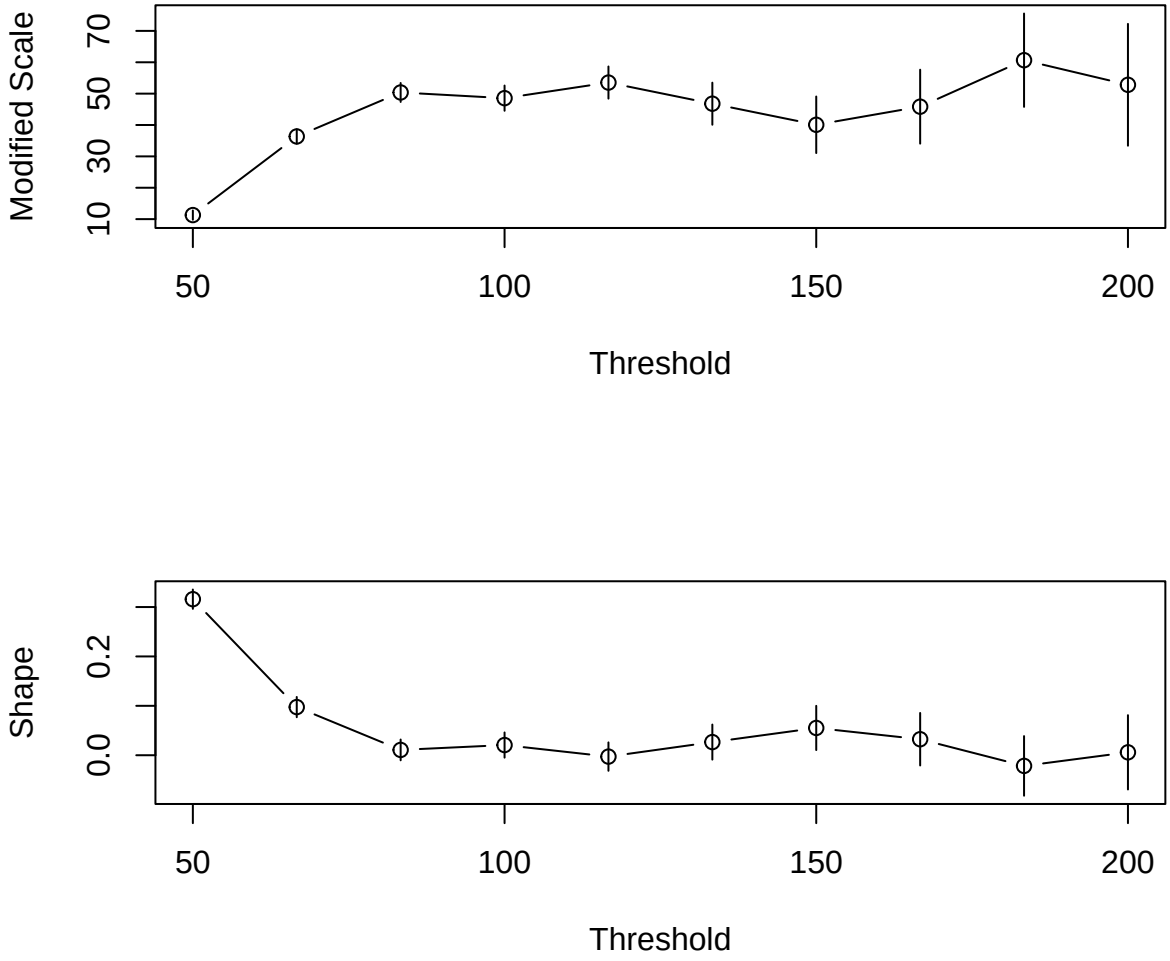


Figure 4: GPD Stability Analysis Plots

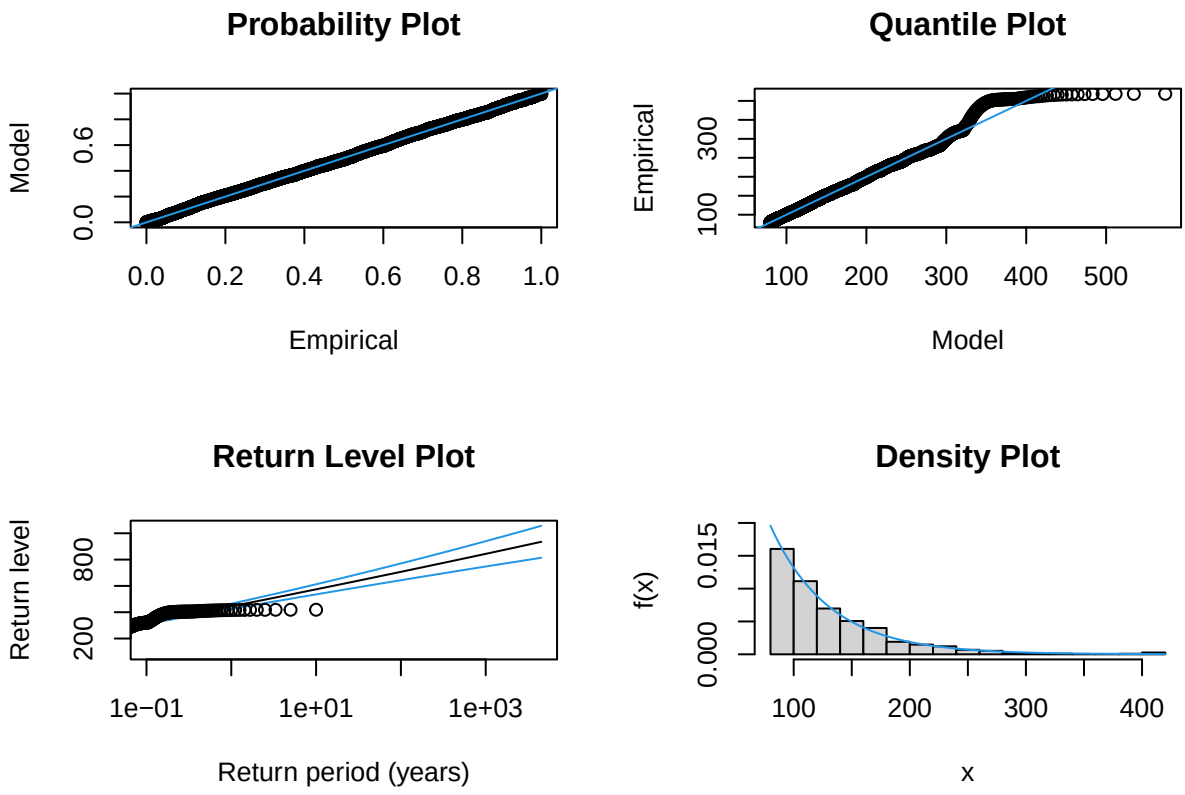


Figure 5: GPD Diagnostic Plots

Independence of Peaks

We apply the Robson and Reed (1999) independence rules from the lecture:

1. Two adjacent peaks must be at least three times the average time to rise apart. The time to rise is the time between a trough and its peak, averaged over at least five clean peaks above the threshold.
2. The flow at the trough between two adjacent peaks must be at least two-thirds lower than the first peak, i.e. below one third of it.

Assumptions: - a clean peak means no other candidate peak in the previous 7 days; - the trough is where the rise starts (within 5% of the week's lowest flow).

If two peaks fail either rule, they belong to the same event and only the larger peak is kept.

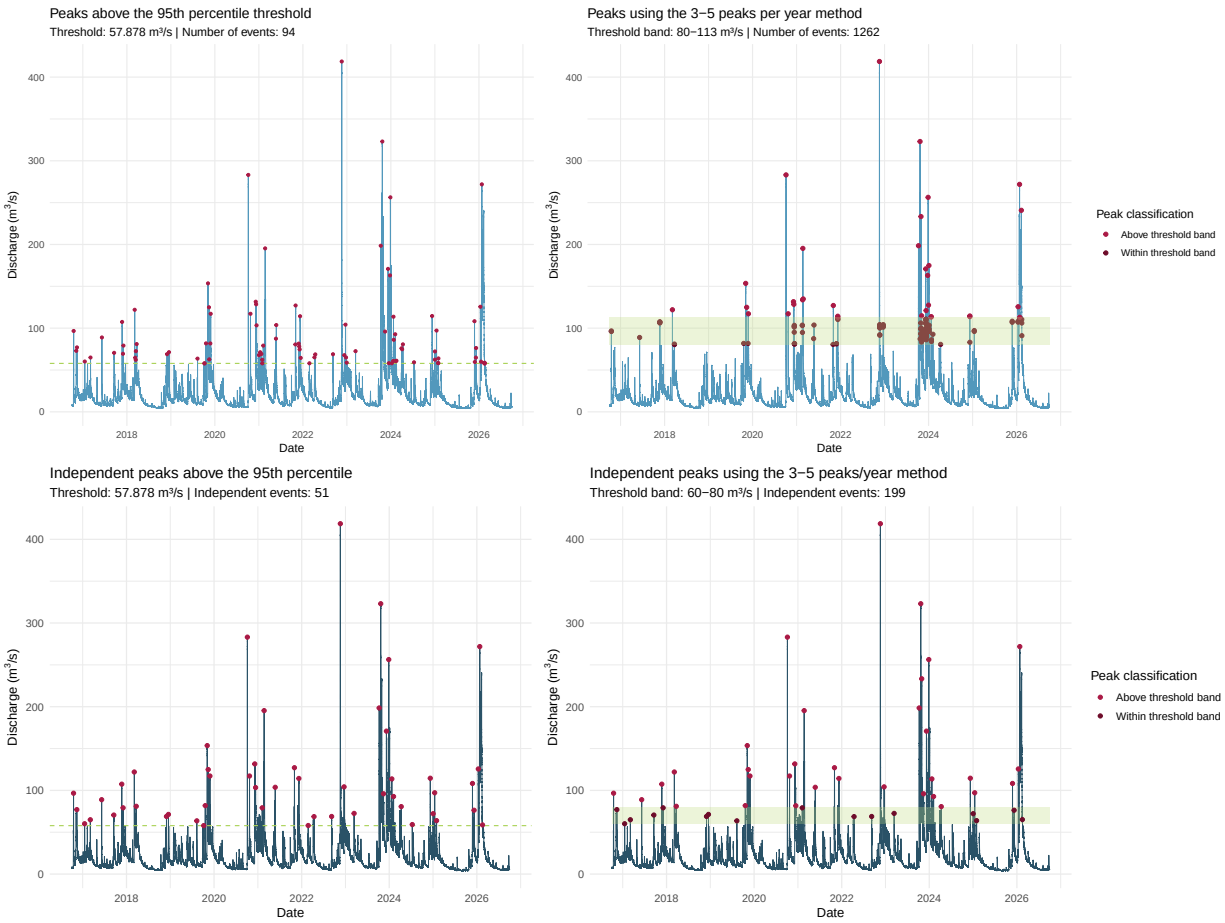
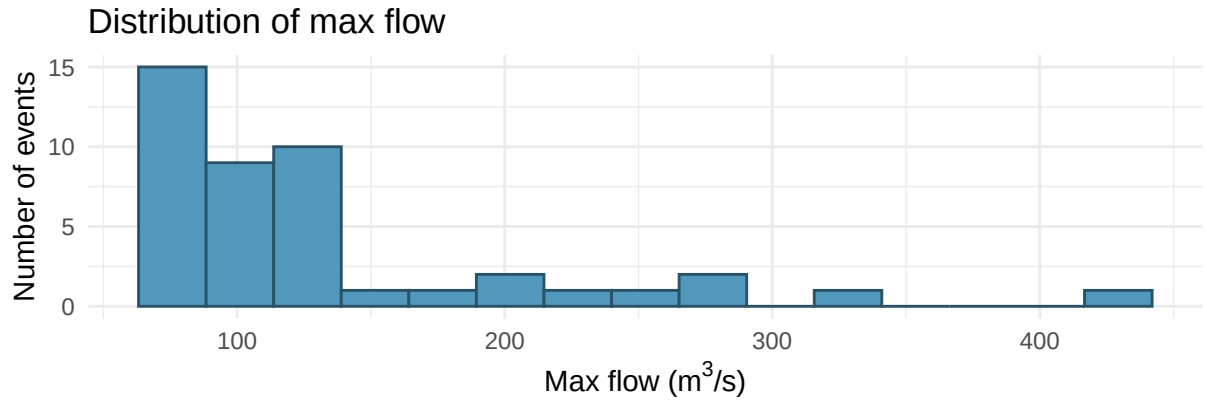
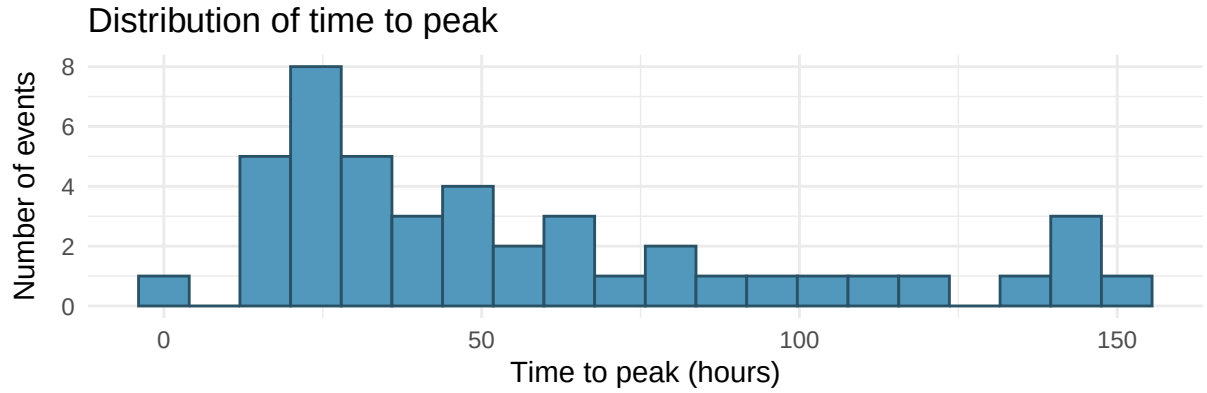


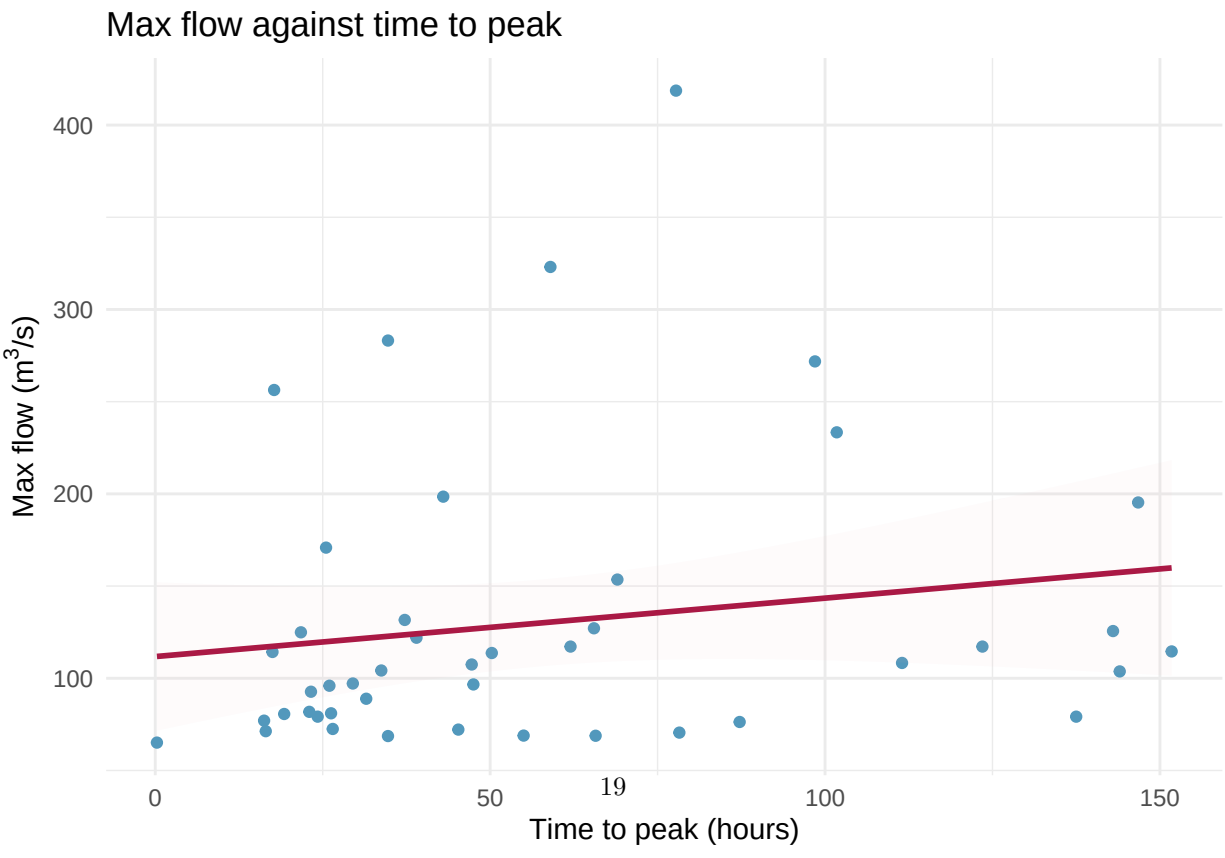
Figure 6: A Comparision of Threshold Methods Prior and Post Independence

Final Threshold Selection

Uncertainty Analysis



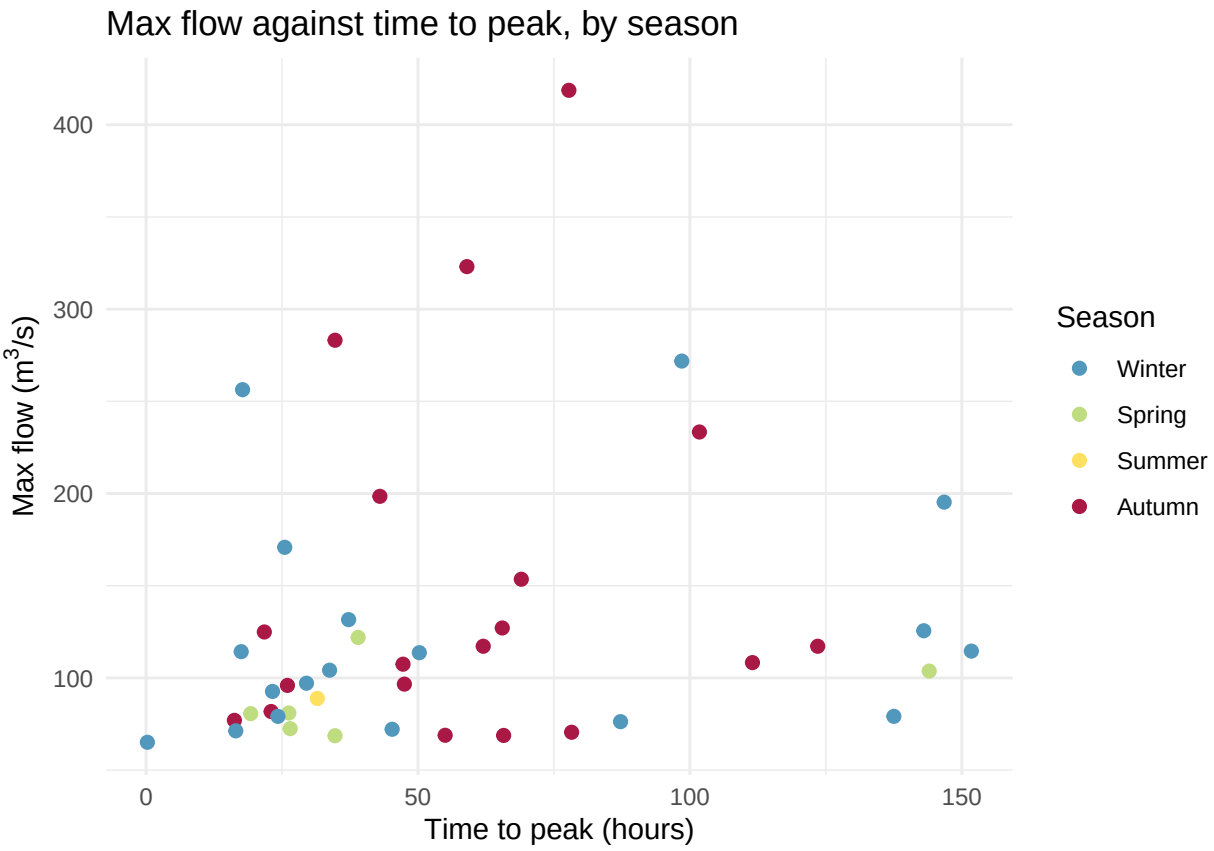


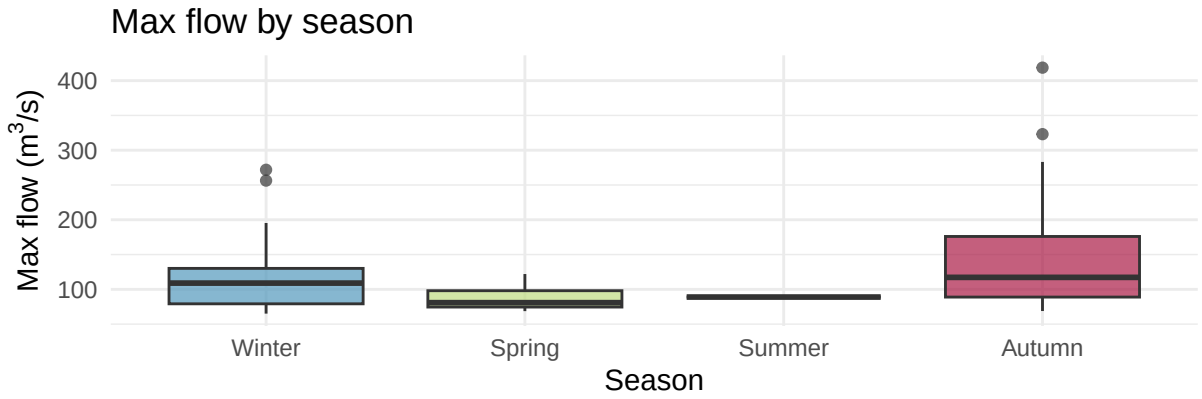


```
##
##  Spearman's rank correlation rho
##
## data:  peaks$time_to_peak_h and peaks$max_flow
## S = 10112, p-value = 0.05852
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.2874159
```

Table 6: Summary of Max Flow and Time to Peak for each Season

season	Events	Median max flow	Median time to peak (h)
Winter	18	108.951	35.500
Spring	6	80.812	30.625
Summer	1	88.889	31.500
Autumn	19	117.185	59.000





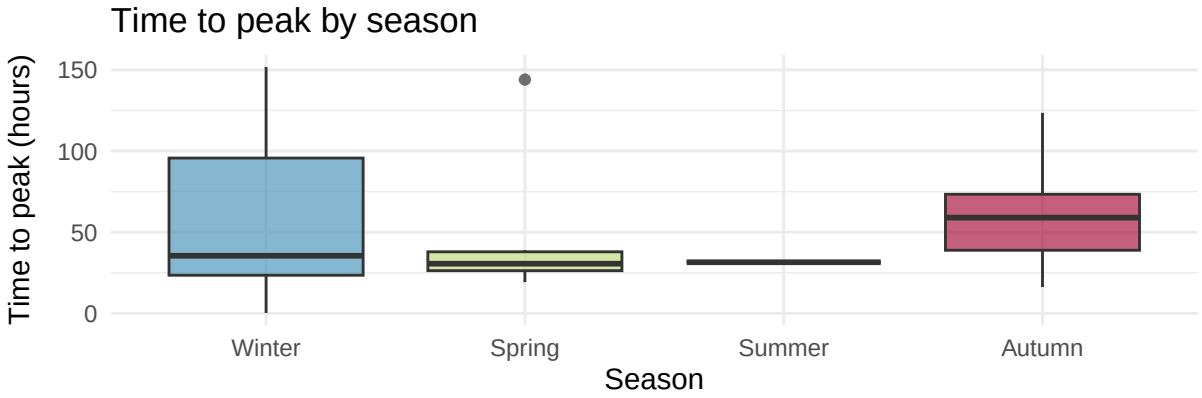


Table 7: Final Uncertanity Table

threshold	events	Events per Year	Median Max Flow	Median Time to Peak (Hours)	Spearman Rho Corelation
55	57	5.7	81.792	43.000	0.066
60	46	4.6	100.434	44.125	0.184
65	44	4.4	105.834	44.125	0.287
70	39	3.9	108.308	43.000	0.270
75	37	3.7	114.256	47.250	0.220